

Power Budget

Team Number: 207

Project Name: Trash Canner

Team Member Names: Gael,Hattie,Riley,Alessandro

Version: 1

All Major Components	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	IR Sensor	TSSP53038	+2 - 5.5V	1	500	500	mA
	Audio Amp	PAM8302AADCR	+2 - 5.5V	1	300	300	mA
	Speaker	CS28-01W75-06-1X	(range)	1	200	200	mA
	Opamp	MCP6004-I/P	+1.8 to 6V	1	100	100	mA
	5V regulator	LM7805T	+12V - 35V	1	1500	1500	mA
+12V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
Regulator	+12V regulator	LM7805T	+12V - 35V	1	1500	1500	mA
	Total Remaining Current Available on +12V Rail					1500	mA

+5V Power Rail	Component Name	Part Number	Supply Voltage Range	Qty.	Absolute Maximum Current (mA)	Total Current (mA)	Unit
	IR Sensor	TSSP53038	+2 - 5.5V	1	500	500	mA
	Audio Amp	PAM8302AADC	+2 - 5.5V	1	300	300	mA
	Speaker	CS28-01W75-06-1X	(range)	1	200	200	mA
	Opamp	MCP6004-I/P	+1.8 to 6V	1	100	100	mA
	Subtotal					1100	mA
	Safety Margin					25%	
	Total Current Required on +5V Rail					1375	mA
Regulator	+5V Regulator	LM7805	+12V - 35V	1	1500	1500	mA
	Total Remaining Current Available on +5V Rail					125	mA

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